

Project Design Phase
Problem – Solution Fit Template

Date	9 th march 2026
Team ID	EL2026TMID10104
Project Name	AI-Based Anomaly Detection Platform
Maximum Marks	2 Marks

Problem – Solution Fit Template:

In many hospitals and healthcare monitoring systems, patient vital signs such as heart rate, oxygen saturation (SpO₂), body temperature, and blood pressure are monitored using traditional systems that rely on fixed threshold values or manual observation. These systems often fail to detect early abnormal patterns in real time.

As a result, health risks may go unnoticed until the patient's condition becomes critical. Delayed detection of abnormal vital signs can lead to serious medical emergencies, delayed treatment, and increased risk for patients.

Purpose

- ☐ To detect abnormal patterns in patient vital signs in real time using artificial intelligence and machine learning techniques.
- ☐ To provide early health risk alerts so that doctors and healthcare staff can respond quickly to critical patient conditions.
- ☐ To improve patient safety by enabling continuous monitoring of vital parameters such as heart rate, oxygen saturation (SpO₂), body temperature, and blood pressure.
- ☐ To support healthcare professionals with a decision-support system that helps identify potential medical emergencies before they become severe.
- ☐ To enhance healthcare monitoring efficiency through real-time data streaming, automated anomaly detection, and instant alert notifications.

Template:

Problem-Solution fit canvas 2.0 – AI-Driven Healthcare Anomaly Detection System

Purpose / Vision:
Real-time Healthcare Monitoring & Early Risk Alerts

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS - Hospitals & Healthcare Providers - ICU & Critical Care Units - Doctors & Healthcare Administrators - Patient Monitoring Centers - Health Tech Companies & Researchers	6. CUSTOMER CONSTRAINTS CC - High cost of continuous monitoring systems - Lack of real-time anomaly detection - Limited access to accurate & timely alerts - Data privacy & security concerns - Complex integration with existing systems	5. AVAILABLE SOLUTIONS AS - Traditional vital sign monitors (static thresholds) - Manual patient observation - Post-incident health records analysis - Basic alert systems (rule-based) - Wearable devices with limited intelligence	Explore AS, differentiate			
	2. JOBS-TO-BE-DONE / PROBLEMS J&P - Monitor patient vitals in real-time - Detect health anomalies early - Reduce delay in critical care - Automate health risk alerts - Improve patient safety & outcomes	9. PROBLEM ROOT CAUSE RC - Delayed detection of health issues - Manual monitoring errors - No real-time analytics on vitals - Overload on healthcare staff - Lack of integrated alert system	7. BEHAVIOUR BE - Doctors rely on dashboards & alerts - Admins monitor critical patients 24/7 - Quick response to high-risk alerts - Trust AI-based decision support - Prefer real-time + email notifications		Focus on J&P, fit into BE, understand RC		
	3. TRIGGERS TR - Abnormal vital signs detected - Patient's health deterioration - Critical event/ Emergency - Scheduled monitoring alerts - Doctor/ Admin login activity	10. YOUR SOLUTION SL - Ai-powered real-time health monitoring system - Anomaly detection using Autoencoder & Isolation Forest - Severity alerts (LOW, MEDIUM, HIGH) - Interactive dashboard + email notifications - Secure database & scalable streaming pipeline	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE - Web Dashboard (Flask) - Email Alerts & Notifications - Admin/Doctor Login Portal 8.2 OFFLINE - Hospital Systems integration - Reports & Data Storage (PostgreSQL) - Emergency Alerts (Email/SMS)			Envision online & offline CH of BE	
	4. EMOTIONS: BEFORE / AFTER EM Before: Stress, uncertainty, delayed action After: Confidence, safety, quick response - Peace of mind, improved efficiency						
	Identify unique TR & EN						

References:

[1] A. Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media, 2019.

[2] F. T. Liu, K. M. Ting, and Z. Zhou, "Isolation Forest," IEEE ICDM, 2008.

[3] Apache Kafka Documentation. <https://kafka.apache.org>